ENVIRONMENTAL RESEARCH
INFRASTRUCTURE AND SUSTAINABILITY

PURPOSE-LED PUBLISHING™ Part of the ENVIRONMENTAL RESEARCH SERIES

PAPER • OPEN ACCESS

Recommendations for cradle-to-gate environmental product declarations (EPD) in 'Buy Clean' procurement based on CDOT's experience

To cite this article: Christopher T Senseney *et al* 2023 *Environ. Res.: Infrastruct. Sustain.* **3** 035004

View the article online for updates and enhancements.

You may also like

- Life cycle assessment of an entirely wood floor system designed for carbon negativity, future adaptability, and end-of-life de/re/construction
Muzan Williams Ijeoma, Sovanroth Ou, Amin Nabati et al.
- A perspective on tools for assessing the building sector's greenhouse gas emissions and beyond
Fiona Greer, Paul Raftery, Gail Brager et al.
- Comparative analysis of life cycle greenhouse gas (GHG) emissions of asphalt from stationary and portable plants: an assessment for Colorado, USA
Daniel Donado-Quintero, Soonhee Han and Christopher Senseney

This content was downloaded from IP address 177.79.97.14 on 01/09/2026 at 18:55

ENVIRONMENTAL RESEARCH
INFRASTRUCTURE AND SUSTAINABILITY

PAPER

Recommendations for cradle-to-gate environmental product declarations (EPD) in 'Buy Clean' procurement based on CDOT's experience

Christopher T Senseney[1,*], **John Harvey**[2], **Ali A Butt**[2] and **Joep Meijer**[3]

[1] Department of Civil, Environmental and Architectural Engineering, University of Colorado, Boulder, CO, United States of America
[2] University of California Pavement Research Center, Department of Civil and Environmental Engineering, University of California, Davis, CA, United States of America
[3] The Right Environment, Boulder, CO, United States of America
[*] Author to whom any correspondence should be addressed.

E-mail: senseney@colorado.edu

Keywords: environmental product declarations, life cycle assessment, civil works, concrete, asphalt, steel, green public procurement

OPEN ACCESS

RECEIVED
27 March 2023

REVISED
2 July 2023

ACCEPTED FOR PUBLICATION
15 August 2023

PUBLISHED
1 September 2023

Original content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence.

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Abstract

The use of environmental product declarations (EPDs) in procurement of construction materials has been of increasing interest to government agencies, and adoption in the United States has been driven in part by 'Buy Clean' legislation. Buy Clean encompasses green procurement policies that promote the purchase of construction materials and products with lower environmental impacts, considering the life-cycle impacts associated with those materials. Most EPDs for transportation infrastructure construction materials are 'cradle-to-gate', or representative of the material production stage of a product and not the entire life cycle; however, these EPDs can allow meaningful quantitative comparisons of environmental and sustainability indicators and serve as a mechanism for measuring improvements in environmental impacts during the production of materials. This paper describes how cradle-to-gate EPDs are used, presents a case study of initial implementation of an EPD policy by the Colorado Department of Transportation, and provides recommendations to address some of the challenges that agencies and stakeholders can face when developing or participating in new EPD programs. It is desired that the recommendations and case study presented in this paper will help provide pathways to agencies for the successful enactment of Buy Clean policies and enable mitigation of actual or perceived barriers to program implementation.

1. Introduction

Procurement of construction materials is one of the processes in the life cycle of transportation infrastructure. Green public procurement (GPP) of construction materials by government agencies, i.e. Buy Clean procurement, can be used as a strategic tool to drive markets into improved environmental outcomes and to help meet government sustainability goals (Bratt *et al* 2013, Sparrevik *et al* 2018, Kundu *et al* 2020, Stokke *et al* 2022). Environmental product declarations (EPD) are gradually being introduced into procurement processes in the U.S. to promote environmental impact considerations. An EPD is a standardized label that resembles the nutrition statement on a food product, presented in a scientifically sound way to communicate the environmental impacts (e.g. resource use, energy, emissions) to either all or part of the life cycle of a product. To produce an EPD, life cycle assessment (LCA) methodology is used in which the life cycle environmental impacts of a material, product, activity, or system are quantified.

When performed in accordance with International Organization for Standardization (ISO) standards for LCA, the product labels are called Type III Environmental Declarations or EPDs under ISO Standard 14025 (ISO 2006a), and the more recently published ISO Standard 21930 (ISO 2017) that was developed specifically for building and transportation infrastructure construction materials. EPDs are produced by following a material's product category rule (PCR), which are generally updated every 5 years. For this paper,

© 2023 The Author(s). Published by IOP Publishing Ltd

'transportation infrastructure' means horizontal construction projects, e.g., roads, bridges, tunnels, drainage, etc. Most EPDs for transportation infrastructure construction materials represent only the material production stage of construction works (Harvey *et al* 2016, Rangelov *et al* 2021), also referred to as cradle-to-gate. Procurement of more environmentally sustainable materials for transportation infrastructure projects can be supported through cradle-to-gate EPDs, which can help decision make informed decisions based on quantified life cycle impact results for the construction materials.

Interest in GPP and Buy Clean policies has grown significantly in recent years, yet there is a gap in the existing literature on GPP and little guidance or program examples for public agencies to follow to implement these types of policies, especially in the U.S. Adopting new procurement policies introduces substantial demands on public sector capabilities and practitioners that can be hampered by a lack of technical capabilities and resources (Uyarra *et al* 2020). In this paper, we present a single case study of the initial launch of a cradle-to-gate EPD policy initiated by the Colorado Department of Transportation (CDOT) in 2022. Case studies can help describe the who, what, where, when, and how of the execution of new policies. Case studies and empirical examples are important for new research areas or areas where the existing theory seems inadequate (Eisenhardt 1989), and case studies are well-suited to help solve contemporary problems and implement policy practices (Yin 2009). It is hoped that the case study presented here will help mitigate actual or perceived barriers to Buy Clean procurement policies and assist others with successful execution of these policies in the future. The primary objectives of this paper are (i) to summarize a case study of the initial implementation of an EPD program for transportation infrastructure projects in Colorado, and (ii) to provide recommendations to agencies and stakeholders interested in using cradle-to-gate EPDs in Buy Clean procurement policies.

2. Background

2.1. Impetus for EPD use

The primary motivation to initially use and develop EPDs in the U.S. was different for the building (vertical) construction industry in comparison to the transportation infrastructure (horizontal) construction industry. The building sector led the way with EPD policies for construction materials, first in Europe in the early 2000s (Braune *et al* 2007), and then in North America in the early 2010s (Gelowitz and McArthur 2016). In the U.S., EPDs were first incentivized by credits from green building rating systems, which led to earlier adoption by the building sector versus the horizontal sector. LEED v4 offered one credit for the submission of 20 EPDs and one more credit for use of products that both have EPDs and have environmental impacts that fall below the industry average (Council US Green Building 2014). An emphasis on producing a certain number of EPDs, as opposed to producing consistent EPDs, may have led to some issues of EPD consistency and harmonization within the building sector (Rangelov *et al* 2021). In the transportation infrastructure sector, development of PCRs and EPDs were a bottom-up effort, initiated by material producers in anticipation of forthcoming green procurement legislation (Mukherjee and Dylla 2017). In many cases, Buy Clean legislation was introduced after PCR directives and EPD guidance were established by construction material producers.

Buy Clean legislation policies that rely heavily on the use of EPDs were introduced in the late 2010s in the U.S. and have expanded to various parts of the country. The Buy Clean California Act (BCCA) was passed in 2017 and requires EPD disclosure for structural steel, concrete reinforcing steel, flat glass, and mineral wool board insulation (California Legislative Organization 2017). The Buy Clean Colorado Act of 2021 requires EPDs for asphalt, concrete, glass, post-tension steel, reinforcing steel, structural steel, and wood structural elements for state funded building projects and asphalt, concrete, and steel for state funded transportation infrastructure projects (Colorado House of Representatives 2021). Buy Clean Oregon, passed in 2022, requires the Oregon Department of Transportation to collect EPDs on asphalt, concrete, and steel (Oregon House of Representatives 2022). In December 2021, the U.S. federal government enacted a Federal Sustainability Plan and Executive Order 14057 to initiate a Federal Buy Clean Initiative to regulate the federal government's purchase of steel, concrete, and flat glass (Executive Order 14057 2021). The federal government also passed the Inflation Reduction Act of 2022 that provided funding to the Environmental Protection Agency, General Services Administration, and Federal Highway Administration to assist producers and manufactures with plans to expand EPD use and validate declarations.

While EPDs can report many other environmental impacts, most Buy Clean legislation specify the collection of global warming potential (GWP) for construction materials and the development of benchmark limits at some future date for maximum allowable GWP per unit of material. According to ISO Standard 21678 (ISO 2020), there are three key benchmark value types: limit (maximum undesired value), reference (current or past practice value), and target (desired practice value). The discussion here focuses on the limit value type of benchmark. Buy Clean legislation often allows for an EPD collection interval or data analysis

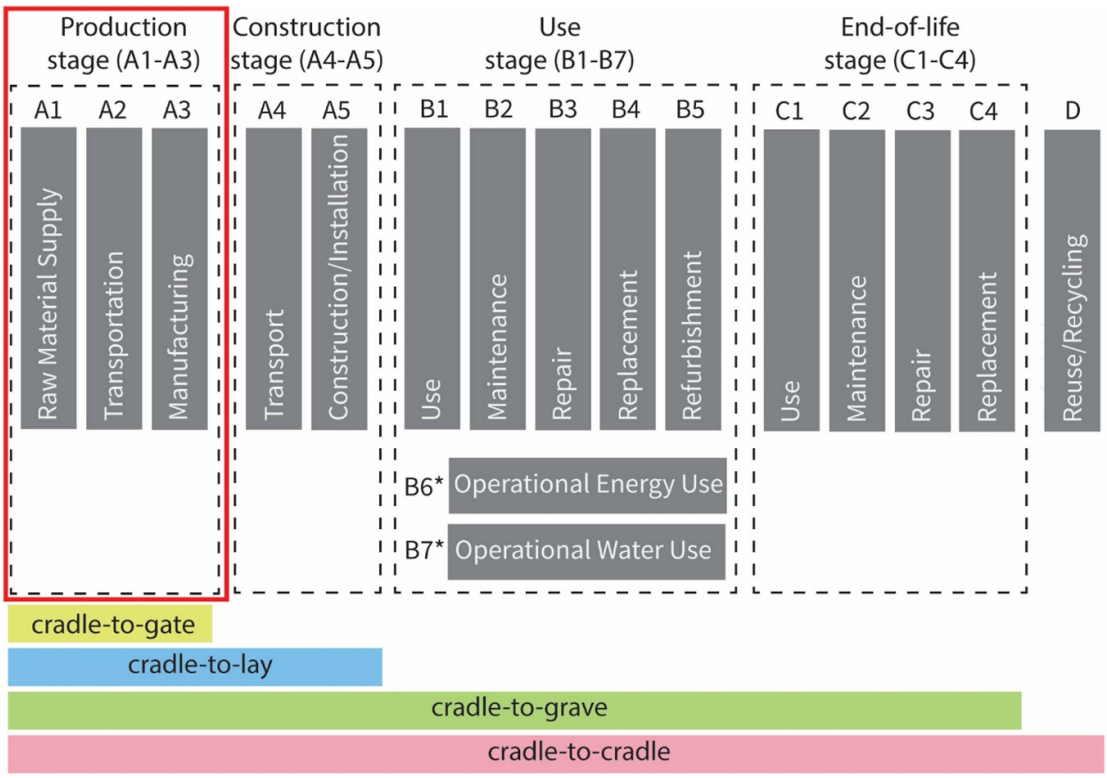


Figure 1. Life cycle stages for construction products and construction works with boundary conditions for different LCA scopes.

period before establishment of limit values. The BCCA was one of the first green public procurement policies to require a state agency to set GWP limits for disclosed EPDs. In compliance with BCCA language, the California Department of General Services set GWP limit values for structural steel, flat glass, and mineral wool board insulation at the national industry average of emissions to manufacture those materials (Department of General Services 2022). The Buy Clean Colorado Act provides CDOT 2.5 years to collect EPD data, and then allows CDOT to use the collected data and nationally recognized EPD databases to develop GWP limits. There have been some studies that provide recommendations for GWP limits for construction materials (Lewis *et al* 2021, Carlisle *et al* 2022, Mattinzioli *et al* 2022); however, literature of the subject of limit setting is sparse.

2.2. What cradle-to-gate EPDs include and when they are submitted

As stated earlier, most EPDs for transportation infrastructure construction materials, especially in the U.S., represent the material production stage of the life cycle of the infrastructure, and not the full life cycle of the material or project. In these cradle-to-gate EPDs, impacts are calculated starting from extraction of raw materials from the earth and ending at the point at which the material/product leaves the gate of the last manufacturing/processing location. Common construction materials for which cradle-to-gate EPDs exist are cement, asphalt mixtures, concrete mixtures, steel, lumber, and aggregates. Figure 1 indicates where cradle-to-gate EPDs fit in the life cycle of a construction project, showing that raw material supply, transport of the raw material within the manufacturing supply chain, and product manufacturing are included. The ISO terminology is presented in figure 1, where modules A1 through A3 are referred to as the 'production stage'.

While it would be preferable to include all life cycle stages for transportation infrastructure materials, cradle-to-gate EPDs have their place and there are valid reasons for these EPDs being limited to modules A1–A3 in Buy Clean policies. ISO 14025 does state that EPDs not based on an LCA covering all life cycle stages have limited comparability (ISO 2006a); however, ISO 14044 also states that omission of life cycle stages can be tolerated in certain EPD applications and should only be allowed if omission does not significantly change the overall outcomes of the study (ISO 2006b). Suppliers of asphalt, concrete, steel, lumber, and aggregates typically have limited control of their product after it leaves the gate; therefore, it is reasonable to pass cradle-to-gate results to contractors, owners, and agencies who have control over and possess more reliable information about the product use and end of life (Santero and Hendry 2016, Rangelov *et al* 2021). To complicate impact assessment during the use phase, the allocation of impacts for in-use

transportation infrastructure construction materials is particularly susceptible to the 'washing machine effect', meaning there is often boundary overlap with other products, e.g., cars, tires, and gasoline (Cullen and Allwood 2009). Others have proposed procedures for assessing the A1–A5 stages of highway pavement projects (Zapata and Gambatese 2005, Bhat *et al* 2020), all life cycle stages of transportation infrastructure construction materials (Trigaux *et al* 2017, Hasan *et al* 2019, Saxe and Kasraian 2020), or cradle-to-cradle sustainability of buildings (European Commision 2022), but such procedures would be difficult to implement in design-bid-build (DBB) contract delivery mechanism predominately used in the U.S. The use and end of life stages for construction materials present further complexity than most manufactured products as these materials are in use for long periods of time and their use can occur in different future social and technical environments than exist today (Gantner *et al* 2018). Based on the current limitations of extending beyond production stage, this study focuses on the application of cradle-to-gate EPDs for Buy Clean policies in the transportation infrastructure construction sector. An analysis of the significance of excluding the life cycle stages after the production stage is beyond the scope of this paper.

The timing of EPD submission is dependent on the type of contact delivery with the vast majority of road, highway, and bridge construction projects in the U.S. utilizing DBB, also known as low bid (Slowey 2018). In DBB project delivery, the sources of the materials to be used on a construction project are usually not known until after the winning bidder (contractor) is selected. Since DBB contractors are only selected based on initial cost (sometimes with consideration of construction duration), cradle-to-gate EPDs are typically submitted to the agency in Buy Clean programs after the project is awarded, when the contractor submits their list of material suppliers or just prior to placement of the material. Material suppliers are usually the entities that develop cradle-to-gate EPDs, and the material suppliers assumed for the bid may not be the actual suppliers used by the contractor during the project. Therefore, environmental impacts (documented in EPDs) of materials used by the contractor that will be delivered after awarding the contract will not typically be part of the contractor selection process. DBB contracts for transportation infrastructure projects are generally either lump sum or unit price. In either case, cradle-to-gate EPDs can be provided to an agency prior to installation of the material. However, if EPDs will be required for the A1–A5 stages, accurate environmental accounting for unit price projects could not be completed until construction is finished, since material quantities and construction schedules are only estimated prior to construction (Bhat *et al* 2020).

Alternative contract delivery mechanisms, such as design-build (DB) and construction manager/general contractor (CM/GC) are sometimes utilized on large, complex transportation infrastructure projects and present unique challenges and opportunities for EPD data collection. For a DB or CM/GC highway pavement construction project, an agency could require the contractor to submit cradle-to-gate EPDs at the time of bidding and then incorporate environmental impacts in procurement decisions as part of best-value procurement as described in (Scott 2006). In this scenario for a DB project, the design build contractor may own the asphalt or concrete plant and house the pavement designer within the firm, which would offer the design build contractor more control in optimizing their bid to minimize environmental impacts. In this scenario for a CM/GC project, the CM would not produce cradle-to-gate EPDs but would likely vet and collect EPDs from material suppliers while developing the proposal during the project pursuit phase. For more information on the roles and responsibilities of different actors in EPD data collection for alternative project delivery contracts, the reader is referred to (Bhat *et al* 2020).

2.3. Types of EPDs

Industry-wide EPDs (IW-EPD), product-specific EPDs (PS-EPD), facility-specific EPDs (FS-EPD), and supply chain-specific EPDs (SCS-EPD) are types of EPDs with different specificity. The definitions presented here for the types of EPDs were chosen by the authors since the terminology associated with EPD types is still quite new and inconsistent in the literature. There is a need for improved terminology alignment on the types of EPDs within the sustainability community. IW-EPD represent typical product impacts from a sector and are not necessarily 'industry-averages' since IW-EPD use production-weighted averages to avoid the risk of one or two small, unusual producers moving the average value to a level which would not be considered typical for the industry (Carlisle *et al* 2022). IW-EPD must also meet representativeness criteria to ensure that upstream processes, geography, and energy data sources have been evaluated and applied consistently. PS-EPD represent the impacts from a single manufacturer for a specific product averaged across multiple facilities. Other literature has referred to PS-EPD as manufacturer-specific EPDs (Ingwersen *et al* 2018). FS-EPD represent the impacts that can be attributed to a single manufacturer at a single manufacturing facility. For example, a FS-EPD for a particular asphalt mixture would reflect the resource use, energy, and emissions from the specific plant where the asphalt is produced. Current PCRs for many civil construction materials encourage the use of PS-EPD and FS-EPD. SCS-EPD is a relatively new term that has evolved with Buy Clean policies and these EPDs represent the highest level of specificity of a material. In addition to being facility specific, SCS-EPD provide primary data for key upstream inputs that have a significant contribution

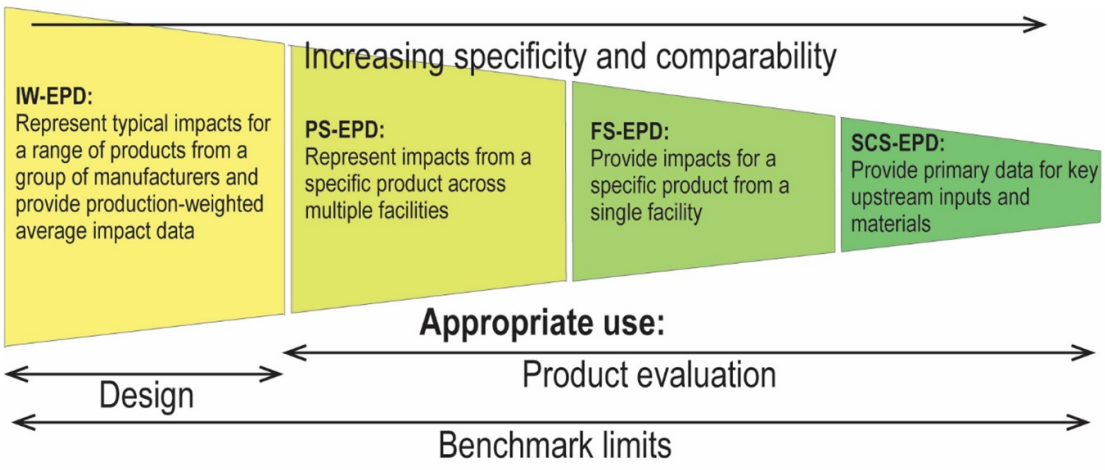


Figure 2. EPD types with appropriate use and increasing levels of specificity and comparability.

to the environmental impacts, usually in the context of GWP and not all environmental impacts. For example, a SCS-EPD for concrete would include facility-specific cement GWP data from the cement plant sourced by a ready-mix supplier rather than industry-wide GWP data that represents an average of nationwide cement manufacturing plants. The 2022 American Center for LCA (ACLCA) PCR Guidance (Bhat *et al* 2022) states that SCS-EPD (although the document does not use the SCS-EPD term) should require facility-specific data for upstream unit processes that cumulatively contribute 50% or more to the disclosed to GWP.

The different types of EPDs offer different levels of comparability and usefulness to decision makers. Early in the design stage where material type selections are made, IW-EPD can be useful in design decisions where understanding the typical impacts of a product might be important, but the exact suppliers and product raw materials are unknown. In procurement, IW-EPD can be considered when determining GWP limit values for materials that are nationally sourced for civil infrastructure, such as hot rolled structural steel, where the trade association-developed IW-EPD represents 90% of the North American production and the GWP values of the different steel mills do not vary significantly (Carlisle *et al* 2022). However, in the procurement stage of most infrastructure construction materials (e.g. asphalt mixtures, concrete, and aggregates), where decisions makers are choosing between different manufactures and products within a given material type, it is more informative and beneficial to use manufacturer- and product-specific data (Waldman *et al* 2020). PS-EPD, FS-EPD, and SCS-EPD are also more appropriate than IW-EPD for use by agencies to establish GWP limit values for construction materials whose impacts vary by region and state because of differences in state specifications, electricity production, methods of extraction for raw materials, material processing methods, and transportation modes and distances. When agencies eventually establish GWP limits, producers should be required to provide PS-EPD, FS-EPD, and/or SCS-EPD during product evaluation, since these EPDs provide the greatest comparability between each other for the same material. Figure 2 shows EPD types with increasing levels of specificity and comparability, and the appropriate use of each of these types.

3. Case study of CDOT's EPD policy implementation for Buy Clean Colorado

A case study of the initial execution of an EPD policy for Buy Clean Colorado by CDOT is presented below. At the time of this paper CDOT had just begun the preliminary launch of their EPD program, where winning contractors had started reporting EPDs for qualifying materials and qualifying projects.

Colorado HB 21-1303, the 'Buy Clean Colorado Act', directs the Office of State Architecture and CDOT to establish policies that reduce GHG emissions over time by accounting for and limiting the GWP of widely used construction materials in state-funded building and transportation projects. The bill passed the state legislature on 7 June 2021, and took effect on 1 July 2022. According to the bill sponsor, State Rep. Tracey Bernett, the goal of the bill is to encourage manufacturers of construction products to reduce their GHG emissions, and ultimately require architects, engineers, and contractors to specify greener construction materials where those materials are practical and economical (Bernett 2022). The Office of the State Architect is responsible for Section 117 of the bill, Colorado Revised Statutes 24-92-117, which covers building construction, and CDOT is responsible for Section 118 of the bill, Colorado Revised Statutes

Table 1. Program-level analysis of 32 primary bid item categories.

	2021	2020	2019	2018	2017	Annual avg. over 5 yrs
All construction bid items excluding services and indirect costs	8404	7290	6991	7240	5468	7078.6
# of 32 primary bid items	3806	3585	3085	3289	2399	3232.8
Total amount of all construction bid items excluding services and indirect costs (M)	$550.3	$484.4	$476.9	$456.2	$325.0	$458.6
Total amount of 32 primary bid items (M)	$404.3	$360.0	$345.3	$336.2	$212.6	$331.7
Percent of 32 primary bid items of all construction bid items excluding services and indirect costs	73.5%	74.3%	72.4%	73.7%	65.4%	72.3%

24-92-118, which covers 'public projects' including roads, highways, and bridges. This case study focuses on the implementation of Section 118, the transportation portion of HB 21-1303, and summarizes the procedures used by CDOT and its advisory team to develop the specifications, protocol, and implementation actions for publicly funded transportation infrastructure projects.

3.1. Development of a protocol

The Buy Clean Colorado Act prescribes asphalt and asphalt mixtures, cement and concrete mixtures, and steel as the eligible materials that require submission of an EPD on CDOT projects (Colorado House of Representatives 2021). CDOT and its advisory team decided the CDOT EPD protocol should specify cradle-to-gate EPDs and should link construction materials that require EPDs to standard CDOT construction bid items. There are 102 bid item categories with a three-digit number and within those categories there are approximately 6,800 items with unique five-digit codes after the three-digit category number (e.g. 403-32601, 412-00615, etc) in the CDOT Bid Item Code Book. The bid item categories are grouped by common materials, e.g., 403 for asphalt mixtures, 412 for concrete pavement, etc.

To establish the EPD protocol to dictate EPD submission that is fair and cost efficient to most stakeholders, CDOT performed multiple quantitative analyses on CDOT bid item expenditures over a 5 year period from 2017–2021, balancing the need for completeness and the need to reflect current or recent practice. Similar bid item analyses have been performed for other agencies to communicate environmental performance and develop cost effective strategies to achieve sustainability goals (Ozer and Al-Qadi 2017). The team obtained bid item data from the publicly available CDOT Engineering Estimates and Market Analysis Project Database. The analyses included all bid items where material is consumed, something is constructed, and/or something is deconstructed, and the analyses excluded all indirect costs (e.g. profit, overhead, contingency, etc) and services like mobilization, surveying, and traffic control.

After reviewing all the items in the CDOT Bid Item Code Book, the team identified 32 three-digit item categories whose primary construction material encompasses the eligible materials from the bill. The bill does not specify which bid items CDOT should be considered for the EPD policy, only that 'additional subcategories within each eligible material' may be established and 'the department shall strive to achieve a continuous reduction of greenhouse gas emissions over time.' As shown in table 1, a 5 year program-level cost analysis determined the 32 primary bid item categories represent an average annual expenditure of $331.7 M out of total annual average of $550.3 M of all construction bid item categories, excluding services and indirect costs from all the values. This program-level analysis indicated that the 32 primary bid item categories represented approximately 72% of the CDOT budget. The 72% value was not designed to meet any specific threshold, but it was simply a check by the team to ensure that a significant or impactful portion of the CDOT materials budget was being analyzed for possible inclusion into the EPD policy.

The team conducted a 95th percentile analysis of the 32 primary bid item categories from 2017–2021 to focus the initial efforts of EPD collection on bid items used frequently in CDOT's policy. It was a goal of the team to not initially ask for EPDs for materials that did not represent a large fraction of CDOT's expenditures. The team somewhat arbitrarily selected the 95th percentile as a cutoff since it produced a natural cut line between heavily utilized materials and lightly utilized materials. As shown in table 2, the team determined the frequency or the number of years (0, 1, 2, 3, 4, or 5 years) that the 32 primary bid item categories fell within the 95th percentile of the sum of the primary bid item expenditures over the 5 year

Table 2. The 95th percentile analysis results of primary bid item categories from 2017 to 2021 with the 13 focus bid items shown in bold.

Primary bid item category	Annual frequency in 95th percentile	Primary bid item category	Annual frequency in 95th percentile
206: Structure Backfill	5	518: Expansion Device	0
310: Full Depth Reclamation	0	**601: Structural Concrete**	5
403: Asphalt Mixtures	5	**602: Reinforcing Steel**	5
405: Heating/Remixing Treatment	0	603: Culverts/Pipes	5
406: Cold Bituminous Pavement	2	**604: Inlets/Manholes**	5
409: Cover Coats	3	**606: Guardrail**	5
411: Asphalt Binder/Emulsion	5	607: Fence	5
412: Concrete Pavement	5	**608: Sidewalk & Bikeway**	4
501: Steel Sheet Piling	0	**609: Curb and Gutter**	4
502: Steel Piling	0	**610: Median Cover**	0
503: Drilled Shaft	5	612: Delineator	0
504: Retaining Wall	5	613: Electrical Conduit	5
507: Slope/Ditch Paving	0	614: Signs	5
509: Structural Steel	1	618: Precast Concrete	4
510: Structural Plate Pipe	0	619: Water Line	1
514: Pedestrian Railing	0	624: Drainage Pipe	1

period. The annual cost of the 32 primary bid item categories were sorted by year from highest to lowest and if the bid item was part of the 95th percentile sum, then the primary bid item was given credit for one year.

For the CDOT 7-1-2022 EPD Protocol, published as Appendix O to the 2023 CDOT Field Materials Manual (FMM) (CDOT 2023), the team narrowed the 32 primary bid item categories down to 13 focus bid item categories (shown in bold in table 2). The general criteria for selecting the 13 focus bid items were frequency of appearance in the 95th percentile analysis, current industry EPD readiness considerations, and functional alignment with the bill. The team felt that these criteria will best allow CDOT to meet the bill requirement to 'achieve a continuous reduction of greenhouse gas emissions over time' in the CDOT EPD policy. Bid items categories 206, 403, 412, 503, 504, 601, 602, 604, 606, 608, and 609 were all included in the 13 focus items since they showed up four or more times in the 95th percentile analysis and are functionally aligned with the bill categories. Bid item categories 310 Full Depth Reclamation and 610 Median Cover did not meet the 95th percentile criteria but were more functionally aligned with the bill than other items, and therefore, were included in the 13 focus items. Bid item categories 603 Culverts/Pipe, 607 Fence, 613 Electrical Conduit, 614 Signs, and 618 Precast Concrete had a frequency of four or more years but were not incorporated in the original 13 focus items since local material manufacturers were not ready to develop FS-EPD for these items.

For the second version of the protocol document, the CDOT 7-1-2023 EPD Protocol, CDOT added new items that required submission of EPDs. Bid items 502 Steel Piling and 509 Structural Steel were added for functional alignment with the bill, improved readiness by steel material manufacturers to produce FS-EPD, and upcoming updates to the Steel PCR that will provide more guidance to manufacturers to develop FS-EPD. Bid Items 603 Culvert/Pipe, 618 Precast Concrete, and 624 Drainage Pipe were added based on frequency of appearance in the 95th percentile analysis, combined with outreach by CDOT to key precast concrete trade groups to help improve EPD readiness of precast suppliers. Bid Item 507 Slope/Ditch Paving was added for functional alignment with the bill. The new protocol also states that uncertainty factors may be applied to GWP values for EPDs that are based on industry average data, i.e. IW-EPD, as a potential penalty for not providing facility specific data.

3.2. Project cost and quantity thresholds

With the aim of not putting excessive burden on small contractors working on small projects and for cost efficiency considerations, the current protocol provided total project cost limit thresholds for EPD submission. In establishing the project cost limits, the team analyzed projects in the 5 year window from 2017 to 2021 with total bids over $1 M and total bids over $3 M. With the $1 M and $3 M project thresholds, 98.9% and 92.3% of total program expenditures of the 13 focus bid items, respectively, would be captured. Accordingly, the team decided on a $3 M total project cost threshold for EPD submission requirements where the total cost would be the final engineer's estimate of the bid items, excluding construction engineering, force account items, and indirect costs. The $3 M project limit can be reduced in the future as the program matures and industries gain EPD development and submission experience.

Similarly, to keep relative EPD development costs to a minimum, the current protocol defined construction material quantity limit thresholds for small quantity exemptions. The CDOT FMM

Table 3. Bid item quantity minimums for EPD submission.

Bid item category	Description	Quantity limits	Units
206	Structure Backfill (Flow-fill)	50	Cubic yard
310	Hydraulic Cement	150	US ton
403	Asphalt Mixtures (HMA/SMA/WMA)	500	US ton
412	Portland Cement Concrete Pavement (PCCP)	1000	Square yare
601	Structural Concrete (Each Class)	50	Cubic yard
602	Reinforcing Steel	15 000	Pound
608	Concrete Sidewalk & Bikeway	250	Square yard
608	Bituminous Sidewalk & Bikeway	500	US ton
609	Curb and Gutter	1000	Linear foot
610	Median Cover Material	4000	Square foot

(CDOT 2023) has a Sampling and Testing of Small Quantities section within the Frequency Guide Schedule that sets small quantity limits for bid item categories 206, 403, 411, 412, 601, and 609, which the team decided to mirror for EPD submission requirements. These quantity limits are generally associated with item cost estimates in the $25 k and $50 k range, and the team developed similar quantity limits in the $25 k to $50 k cost estimate range for the other focus bid items without small quantity limits in the FMM. The FMM quantity limits for Item 609 were also modified to match the $25 k to $50 k range. The quantity limits for ten of the bid item categories in the CDOT EPD Protocol are shown in table 3.

3.3. Implementation timeline

The CDOT EPD Protocol applies to all projects advertised on or after 1 July 2022. Partly because both concrete and asphalt pavement projects in rural areas of Colorado often employ portable batch plants where the source of aggregates and other feedstock materials may be unknown at the time of bidding, the protocol allowed for EPD submission of eligible materials a minimum of two weeks prior to materials placement, or before they are permanently incorporated into the work.

The initial phase of the CDOT EPD program involves collecting baseline data for construction materials to establish threshold limits from the collected data. To obtain Colorado regional specific data to be used by CDOT to develop future GWP benchmark limits, CDOT is requesting that contractors submit FS-EPD or PS-EPD as opposed to IW-EPD for the initial data collection. By 1 January 2025, CDOT must establish a policy with GWP limits on eligible materials. CDOT may create subset limits with the eligible material categories (e.g. different limits for different strength concrete mixtures). By the 1 July 2025, the winning contractor will be required to submit EPDs for eligible materials and those EPDs must comply with maximum GWP limits established by the CDOT policy. At the time of this paper, CDOT is gathering EPD data and has not determined how the GWP threshold limits will be determined. Ongoing research and analysis are investigating the most appropriate methods to set threshold limits for each regulated material type and how the thresholds will be implemented and enforced.

4. Discussion

There are general lessons that can be derived from Colorado's and other U.S. states' implementation of Buy Clean EPD programs for transportation infrastructure construction materials. When developing future public procurement policies for construction materials, agencies, legislator, and stakeholders should consider the following recommendations:

a. Buy Clean legislation should consider separating the transportation component, i.e. horizontal construction, and building component, i.e. vertical construction, oversight responsibilities of EPD policies. Ideally, the horizontal construction component should be managed by the state highway agency, and the vertical construction component should be managed by the division of the state architect, or equivalent. Construction materials used on highway projects, such as asphalt, can vary significantly from the same materials used in the building construction sector, and materials supply and production can be different, such as portable plants on rural highway projects.
b. Buy Clean EPD policies should require the use of regionally based thresholds based on PS-EPD, FS-EPD, and SCS-EPD collected from the pool of suppliers to the agency. Although SCS-EPD do not currently represent the majority of EPDs, the ideal state is one where SCS-EPD are required for limit-setting and compliance.

c. EPD programs should balance the number of product categories and material types with the complexity of setting up categories and managing them. Programs should focus on material categories with the largest use and largest impacts.
d. The variability of EPDs can be high and caused by numerous factors, including choices regarding background data (Butt and Harvey 2021). EPD programs should adopt approaches that reduce variability of EPD results for a given material, including more prescriptive PCRs and standardized use of regional background data.
e. It is important for agencies to identify whether materials being characterized for LCA, compared through cradle-to-gate EPDs for use in design, or evaluated for procurement will have the same performance throughout the rest of the life cycle. This means that the materials being compared should provide the same functionality, meaning that they should have equivalent functionality over comparable time periods and life cycle stages.
f. There is need for improved terminology alignment for the definitions of the different types of EPDs (e.g. IW-EPD, PS-EPD, FS-EPD, SCS-EPD).
g. National standards should be established for electronic reporting format, units, and other presentation of information in EPDs, that can be incorporated into PCRs. There should also be agreed upon standards for EPD data quality. The 2022 ACLCA PCR Guidance provides some frameworks for producing standard conformant and consistent PCRs and includes recommendations for the digital transfer of EPD data (Bhat *et al* 2022).

5. Conclusion

Use of cradle-to-gate EPDs can be an important part of the implementation of life cycle thinking to improve the sustainability of transportation infrastructure. The use of EPDs in procurement of transportation infrastructure construction materials has been of increasing interest to government agencies, and implementation has been driven in part by Buy Clean legislation. However, EPDs need to be specified and managed well to make them an efficient and reliable tool for helping to improve environmental sustainability. This paper reviewed the use of cradle-to-gate EPDs for construction materials, presented a case study of the initial implementation of an EPD policy in Colorado, and offers recommendations to address the challenges that agencies and industry stakeholders may face when developing and participating in EPD programs. It is hoped that the case study and recommendations presented here will spur advancement of GPP in more parts of the U.S. Future research should consider the expansion of Buy Clean procurement polies to all life cycle stages, and the development of guidelines for benchmarking limits and the impact of those limits on the different contract delivery mechanisms for construction projects.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

Acknowledgments

This work was partially funded by the Colorado Department of Transportation (CDOT) through Agreement Number 22-HTD-ZH-00013. The views and opinions expressed in this paper are those of the authors and do not necessarily reflect the views or positions of CDOT or its employees.

ORCID iDs

Christopher T Senseney https://orcid.org/0000-0001-9709-1268
John Harvey https://orcid.org/0000-0002-8924-6212
Ali A Butt https://orcid.org/0000-0002-4270-8993

References

Bernett T 2022 History of HB-1303 Personal interview, Jun 1 (Unpublished)
Bhat C G, Shacat J and Mukherjee A 2020 Technical and organizational challenges to developing product category rules for asphalt pavement construction *InPavement, Roadway, and Bridge Life Cycle Assessment 2020* (CRC Press) pp 100–10
Bhat C G, Tjan A, Mellantine J, Feraldi R, Lasso A, Swack T, Mukherjee A, Dylla H and Randglov M 2022 ACLCA PCR guidance—process and methods toolkit *American Center for Life Cycle Assessment*
Bratt C, Hallstedt S, Robèrt K H, Broman G and Oldmark J 2013 Assessment of criteria development for public procurement from a strategic sustainability perspective *J. Clean. Prod.* **52** 309–16

Braune A, Kreissig J and Sedlbauer K 2007 The use of EPDs in building assessment-towards the complete picture *Portugal SB07-Sustainable Construction, Materials and Practices, Challenge of the industry for the new milenium, PTS* 1 pp 299–304
Butt A A and Harvey J 2021 Lessons learned from Caltrans pilot program for implementation of EPDs *Report UCPRC-TM-2021-01* (California Department of Transportation, prepared by University of California Pavement Research Center)
California Legislative Organization 2017 Buy Clean California Act (3500–3505)
Carlisle S, Waldman B, DeRousseau M, Miller L, Ciavola B, Lewis M and Simonen K 2022 Buy clean California limits: a proposed methodology for assigning industry-average GWP values for steel, mineral wool, and flat glass in California *Carbon Leadership Forum* (University of Washington)
CDOT 2023 Field materials manual. Colorado department of transportation
Colorado House of Representatives 2021 *House Bill 21–1303 "Buy Clean Colorado Act". Seventy-third General Assembly* (Colorado State Legislature)
Council US Green Building 2014 LEED v4 for building design and construction USGBC Inc [Preprint]
Cullen J M and Allwood J M 2009 The role of washing machines in life cycle assessment studies: the dangers of using LCA for prioritization *J. Ind. Ecol.* **13** 27–37
Department of General Services 2022 *Buy Clean California Act Legislative Report*
Eisenhardt K M 1989 Building theories from case study research *Acad. Manage. Rev.* **14** 532–50
European Commision 2022 *Level(s): European Framework for Sustainable Buildings* (Brussels)
Executive Order 14057 2021 Catalyzing clean energy industries and jobs through federal sustainability (available at: www.federalregister.gov/documents/2021/12/13/2021-27114/catalyzing-clean-energy-industries-and-jobs-through-federal-sustainability) (Accessed 9 Jun 2023)
Gantner J, Fawcett W and Ellingham I 2018 Probabilistic approaches to the measurement of embodied carbon in buildings *Embodied Carbon in Buildings: Measurement, Management, and Mitigation* pp 23–50
Gelowitz M D and McArthur J J 2016 Investigating the effect of environmental product declaration adoption in LEED® on the construction industry: a case study *Procedia Eng.* **145** 58–65
Harvey J, Meijer J, Ozer H, Al-Qadi I L, Saboori A and Kendall A 2016 Pavement life cycle assessment framework *Federal Highway Administration* (United States)
Hasan U, Whyte A and Al Jassmi H 2019 Critical review and methodological issues in integrated life-cycle analysis on road networks *J. Clean. Prod.* **206** 541–58
Ingwersen W, Zeman J, Lasso A, Pearson C and Montalbo T The ACLCA PCR guidance document: an overview and future plans *Proc. of LCA XVIII, "Connect, Learn, Transform" (Fort Collins, CO, 25–27 September 2018)* pp 117–24
ISO 2017 ISO 21930:2017: sustainability in buildings and civil engineering works—core rules for environmental product declarations of construction products and services
ISO 2020 ISO 21678:2020—Sustainability in Buildings and Civil Engineering Works—Indicators and benchmarks—Principles, Requirements and Guidelines
ISO. ISO 2006a 14025:2006: environmental labels and declarations—type III environmental declarations—principles and procedures
ISO. ISO 2006b 14044:2006: environmental management—life cycle assessment—requirements and guidelines
Kundu O, James A D and Rigby J 2020 Public procurement and innovation: a systematic literature review *Sci. Public Policy* **47** 490–502
Lewis M, Huang M, Waldman B, Carlisle S and Simonen K 2021 *Environmental Product Declaration Requirements in Procurement Policies* (InCarbon Leadership Forum. University of Washington)
Mattinzioli T, Presti D L and Del Barco Carrión A J 2022 A critical review of life cycle assessment benchmarking methodologies for construction materials *Sustain. Mater. Technol.* **33** e00496
Mukherjee A and Dylla H 2017 Challenges to using environmental product declarations in communicating life-cycle assessment results: case of the asphalt industry *Transp. Res. Rec.* **2639** 84–92
Oregon House of Representatives 2022 *House Bill 4139. 81st Oregon Legislative Assembly*
Ozer H and Al-Qadi I L 2017 Regional LCA tool development and applications *Pavement LCA Symp. (12 April)*
Rangelov M, Dylla H, Mukherjee A and Sivaneswaran N 2021 Use of environmental product declarations (EPDs) of pavement materials in the United States of America (USA) to ensure environmental impact reductions *J. Clean. Prod.* **283** 124619
Santero N and Hendry J 2016 Harmonization of LCA methodologies for the metal and mining industry *Int. J. Life Cycle Assess.* **21** 1543–53
Saxe S and Kasraian D 2020 Rethinking environmental LCA life stages for transport infrastructure to facilitate holistic assessment *J. Ind. Ecol.* **24** 1031–46
Scott S 2006 NCHRP Report 561: best-value procurement methods for highway construction projects *Transportation Research Board*
Slowey K Construction Dive 2018 The dotted line: why is design-bid-build still the No. 1 process? (Industry Dive) (available at: www.constructiondive.com/news/the-dotted-line-why-is-design-bid-build-still-the-no-1-process/517308/)
Sparrevik M, Wangen H F, Fet A M and De Boer L 2018 Green public procurement–A case study of an innovative building project in Norway *J. Clean. Prod.* **188** 879–87
Stokke R, Kristoffersen F S, Stamland M, Holmen E, Hamdan H and De Boer L 2022 The role of green public procurement in enabling low-carbon cement with CCS: an innovation ecosystem perspective *J. Clean. Prod.* **363** 132451
Trigaux D, Wijnants L, De Troyer F and Allacker K 2017 Life cycle assessment and life cycle costing of road infrastructure in residential neighbourhoods *Int. J. Life Cycle Assess.* **22** 938–51
Uyarra E, Zabala-Iturriagagoitia J M, Flanagan K and Magro E 2020 Public procurement, innovation and industrial policy: rationales, roles, capabilities and implementation *Res. Policy* **49** 103844
Waldman B, Huang M and Simonen K 2020 Embodied carbon in construction materials: a framework for quantifying data quality in EPDs *Build. Cities* **1** 625–36
Yin R K 2009 *Case Study Research: Design and Methods* 4th (Sage)
Zapata P and Gambatese J 2005 Energy consumption of asphalt and reinforced concrete pavement materials and construction *J. Infrastruct. Syst.* **11** 9–20